INTRODUCTION TO HYBRID AND ELECTRIC VEHICLES

Subject Code : 23A0360Se

UNIT IV

Electric and Hybrid Vehicles - Case Studies

I. Series and Parallel Hybrids - Operating Strategies

The case studies in this unit are based on the two basic hybrid architectures (series, parallel) and their combinations. Operation is described by two strategies: **charge sustaining** and **charge depleting**.

Series hybrid

- Engine -> generator -> electric motor; battery buffers power. Wheels driven only by the motor. Ex: Chevrolet Volt (in extended range), diesel-electric locomotives and hybrid buses.

Parallel hybrid

- Engine and motor both couple mechanically to wheels. Ex: Honda Insight / Civic hybrid (IMA), many mild hybrids.

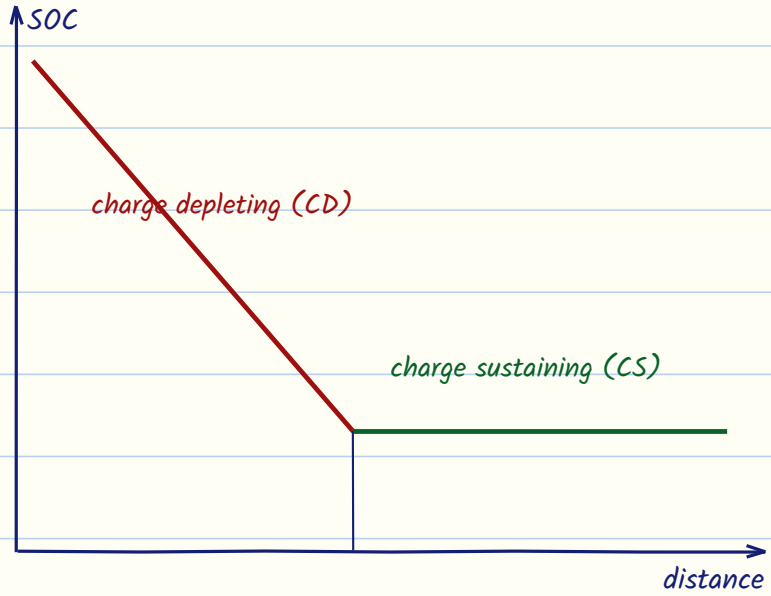
Charge sustaining (CS) mode

- The battery SOC is **maintained within a narrow band** about a target (say 50-60 %). The engine supplies the average power; the battery only supplies peaks and absorbs

regenerative energy. Net battery energy change over the trip is about zero; all energy comes from fuel. Used in conventional HEVs such as the Prius and Insight.

Charge depleting (CD) mode

- The vehicle uses **stored battery energy** (from the grid) to propel, SOC falls from high to a lower limit. This is the **all-electric range (AER)** of a PHEV / EREV. After reaching the lower SOC limit the vehicle switches to CS mode (Volt: about 40 km EV, then engine runs).

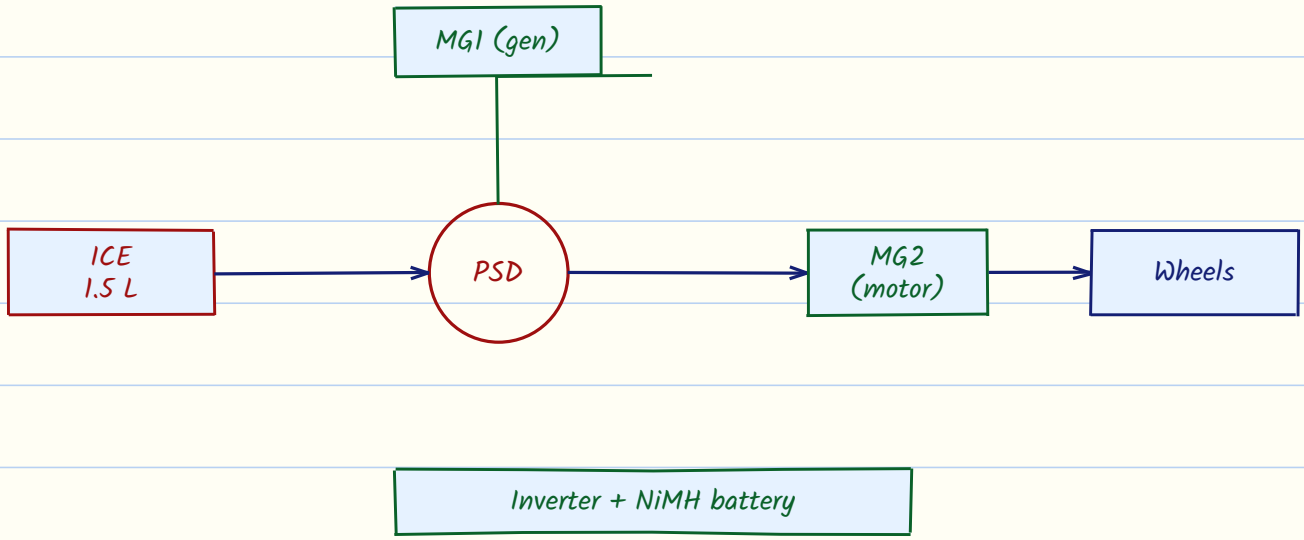


CD : battery energy used up, SOC falls (EV mode).

CS : SOC held near a set value; fuel supplies all the energy.

2. Hybrid Vehicle Case Study - Toyota Prius

The **Toyota Prius** (1997 Japan, 2000 world) is the world's first mass-produced hybrid and best-selling HEV. It is a **series-parallel (power-split)** hybrid with the **Toyota Hybrid System (THS / HSD)**.



- **Engine**: Atkinson-cycle petrol engine - 1.5 L, about 57 kW (2nd gen); later 1.8 L, about 73 kW. Atkinson cycle gives high efficiency but lower power density, compensated by the motor.
- **Power split device**: planetary gear. Engine -> planet carrier, MG1 (generator) -> sun gear, MG2 (traction motor) and wheels -> ring gear. Acts like an **electronic CVT**; no clutch, no conventional gearbox.
- **Motors**: MG1 and MG2 are PM synchronous (AC) machines; MG2 about 50 kW (2nd gen), 60 kW (3rd gen), fed by an inverter; boost converter raises battery voltage (201.6 V -> up to 500 V) in later models.
- **Battery**: sealed NiMH, about 201.6 V (168 cells in 2nd gen), about 6.5 Ah; SOC kept between 40-80 %.
- **Modes**: starts and low speed on motor only; normal driving

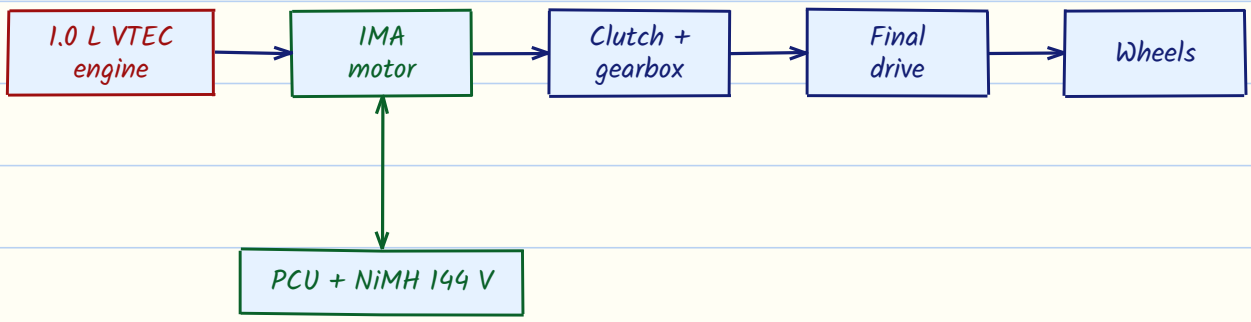
engine + MG2, MG1 charges the battery; full throttle engine + battery power; braking - regeneration by MG2 (+ hydraulic brake blending); engine auto stop at standstill.

- Fuel economy about 4-5 L/100 km (about 20 km/L); emissions much lower than comparable ICE cars.

3. Hybrid Vehicle Case Study - Honda Insight

The first-generation **Honda Insight (1999)** is a two-seat **parallel mild hybrid** using the **Integrated Motor Assist (IMA)** system. The thin permanent-magnet DC brushless motor is mounted directly on the engine crankshaft, between engine and transmission, so no separate clutch for it.

Motor sits between engine and transmission (IMA)



- **Engine**: 1.0 L, 3-cylinder, VTEC-E lean burn, about 50 kW (67 hp). **Motor**: about 10 kW (13 hp) assist; works as starter, assist and generator (regeneration).
- **Battery**: NiMH, 144 V, 6.5 Ah (120 cells) under the rear

floor. Transmission: 5-speed manual or CVT.

- **Light weight** (aluminium body about 850 kg) and excellent aerodynamics (C_D = 0.25) help to give about 3.4 L/100 km (very high economy).
- Modes: motor assists at acceleration; cruise by engine alone; regeneration on deceleration; **idle stop**; lean-burn cylinder deactivation.
- Merits: simple, cheap, only one electric machine; demerit: cannot run in pure-EV mode, assist motor is small.

4. Hybrid Vehicle Case Study - Chevrolet Volt

The **Chevrolet Volt (2010-11)** is a **plug-in extended-range electric vehicle (EREV)**. It is mostly an electric car with a small petrol engine used only for generating power (or assisting at high speed).

Chevrolet Volt operating modes

1. EV mode (CD)	battery -> motor -> wheels; engine off
2. Extended range (CS)	engine -> generator -> motor; battery holds SOC
3. High-speed assist	engine also helps drive through planetary gears
4. Regeneration	motor -> battery while braking

- **Battery**: T-shaped Li-ion pack, about 16 kWh (about 10.5 kWh usable), 288 cells, liquid-cooled; **EV range about 35-40 miles (about 60 km)**; charged from a 120 / 240 V

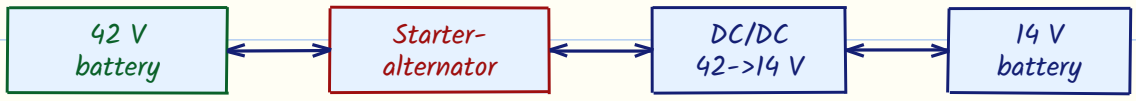
outlet in about 4-10 hours.

- **Motors**: main traction motor about 111 kW (150 hp), 370 N-m; second motor/generator about 55 kW. Engine: 1.4 L petrol, about 63 kW. Planetary gear and clutches allow engine to assist directly at higher speed (2nd gen improved).
- Total range about 600 km. Charge depleting first, then charge sustaining. Typical commuter drives entirely on grid electricity.

5. 42 V System for Traction Applications

Modern cars need more and more electrical power (steering, air-conditioning, valves, heating, infotainment). The old **14 V (12 V battery)** system can supply only about 2-3 kW because of high current. The industry proposed the **42 V PowerNet** (36 V battery, 42 V charging).

Dual-voltage system: heavy loads on 42 V, lamps / electronics on 14 V



engine start, torque assist, regen

EPS, A/C compressor, valve actuators, heated catalyst on 42 V bus

- Higher voltage -> for the same power the current is 3

times less, so thinner copper wire, smaller connectors, less I^2R loss; allows 5-10 kW generators.

- Enables **starter-alternator (integrated starter generator, ISG / BSG)**: engine start-stop, torque assist, regeneration, electric power steering, electric water pump, electromechanical valve actuators, electric A/C compressor, electrically heated catalyst.
- Issues: arcing at switches above about 30 V (needs solid-state switches), new bulbs / motors, safety below 60 V (not hazardous - below the touch-voltage limit), dual-voltage system complexity. It was later replaced by the **48 V** mild hybrid standard.

6. Lightly Hybridised Vehicles and Low-Voltage Systems

- **Micro hybrid**: only stop-start (belt starter), 12 V; saves 3-8 % fuel.
- **Mild hybrid**: 42 / 48 V system with belt-driven starter-generator (BSG) or crank-mounted ISG of 5-15 kW; provides stop-start, regenerative braking and light **torque assist**; **cannot drive the vehicle by motor alone**. Ex: Honda IMA, GM eAssist, Suzuki SHVS, Maruti Ciaz / Ertiga (SHVS), Mercedes EQ Boost.
- Fuel economy gain: 10-20 %; cost is modest compared to

full hybrids; small battery (0.3-1 kWh, Li-ion or supercap).

- **Low voltage (below 60 V DC)** is regarded as safe for people (no insulation hazard) and cheaper to build.

Type	Voltage	Motor power	Electric-only drive	Fuel saving
Micro	12 V	2-3 kW	No	3-8 %
Mild	42/48 V	5-15 kW	No	10-20 %
Full	200-300 V	30-60 kW	Yes	30-50 %
Plug-in	300+ V	60-110 kW	Yes (long)	50 %+

7. Electric Vehicle Case Studies

(a) GM EV1

- General Motors **EV1 (1996-1999)**, the first modern mass-produced electric car (leased in California / Arizona). Two-seat, very streamlined (C_D = 0.19), aluminium structure.
- **Motor**: 3-phase AC induction motor about 102 kW (137 hp) with IGBT inverter; single-speed reduction gear; regenerative braking.
- **Battery**: Gen 1 lead-acid 16.5 kWh (range 55-95 miles); Gen 2 (1999) NiMH 26.4 kWh (range about 100-140 miles). Top speed limited to 80 mph (about 130 km/h);

0-60 mph in about 8 s. Inductive (Magne Charge) charging.

- Program ended in 2002-03 (leases not renewed); important as the technology demonstrator that inspired Volt and later EVs.

(b) Nissan Leaf

- **Nissan Leaf (2010)** - best-selling BEV for years. 5-door hatchback front-wheel drive.
- **Motor**: 80 kW (107 hp) PM synchronous AC motor, 280 N-m, single-speed reduction.
- **Battery**: 24 kWh (2010) laminated-cell Li-ion pack (48 modules, about 360 V) under floor; range about 117 km EPA (about 175 km NEDC); later 30 / 40 / 62 kWh versions.
- Charging: 3.3 / 6.6 kW AC onboard charger (8 h on 240 V), **CHAdeMO DC fast charge** 80 % in about 30 min. Also supports vehicle-to-home (V2H).

(c) Mitsubishi i-MiEV

- **i-MiEV (2009)** - based on the small "i" kei car; rear-mounted motor and rear-wheel drive; low floor and compact.
- **Motor**: 47 kW PM synchronous, 180 N-m. **Battery**: 16 kWh

Li-ion (88 cells, 330 V) under the floor; range about 160 km (JC08), about 100 km real.

- Charging: 15 h on 100 V, 7 h on 200 V, 80 % in 30 min with CHAdeMO quick charge. Top speed 130 km/h.

Parameter	GM EV1	Nissan Leaf	i-MiEV
Year	1996 / 99	2010	2009
Motor	AC induction, 102 kW	PMSM, 80 kW	PMSM, 47 kW
Battery	Lead-acid / NiMH	Li-ion 24 kWh	Li-ion 16 kWh
Range	~100-225 km	~117-175 km	~100-160 km
Top speed	130 km/h	145 km/h	130 km/h

8. Hybrid Electric Heavy-Duty Vehicles

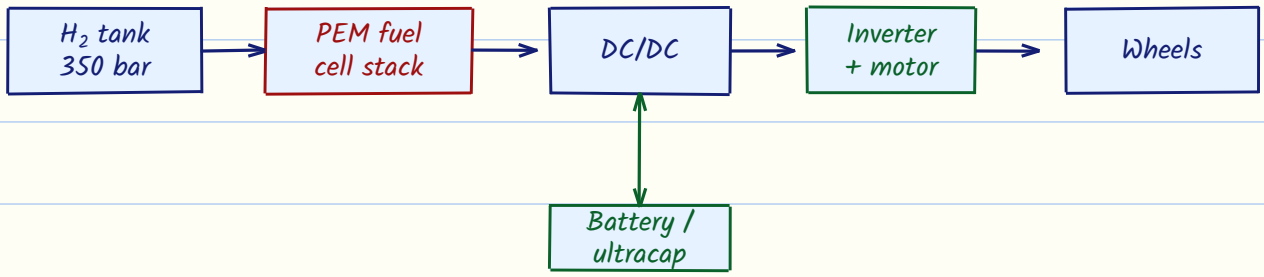
Trucks, transit buses, refuse trucks and military vehicles have stop-go duty cycles and large mass, so **regenerative braking and engine downsizing** give large fuel savings (20-40 %).

- **Series hybrid buses** (Orion VII, Alexander Dennis Enviro 400H): diesel engine drives generator; large battery / ultracapacitor and wheel-hub or central traction motor. Engine runs at constant best-efficiency point.
- **Parallel hybrid trucks** (Eaton, Volvo I-SAM, Kenworth T270): motor-generator between engine and automated transmission; torque assist and regeneration.

- **Hydraulic hybrids** (UPS / refuse trucks) store energy in a pressurised accumulator instead of batteries.
- Challenges: heavy and costly battery, thermal management, high power converters (200-600 V, 100-250 kW), duty cycle dependency; use ultracapacitors for peaks.

9. Fuel Cell Heavy-Duty Vehicles

Fuel-cell hybrid heavy-duty power train



- **PEM fuel cell** (proton exchange membrane): anode $H_2 \rightarrow 2H^+ + 2e^-$; cathode $\frac{1}{2}O_2 + 2H^+ + 2e^- \rightarrow H_2O$. Electric output about 0.6-0.7 V per cell, efficiency 40-60 %; the only exhaust is water.
- Hydrogen stored at 350 / 700 bar in carbon-fibre tanks; refuelling in 5-10 minutes; range 300-500+ km, good for buses and trucks needing long range and heavy payload.
- Fuel cell is slow to respond, so it is **hybridised with a battery / ultracapacitor** for peak power and regeneration. Ex: Ballard-powered city buses, Mercedes Citaro FuelCELL,

Toyota Mirai (cars), Hyundai XCIENT Fuel Cell truck.

- + Zero tail-pipe emission, long range, fast refuelling. - High cost (platinum catalyst), need hydrogen production / distribution, durability and cold-start issues.

Important questions: (1) Charge sustaining vs charge depleting. (2) Toyota Prius / Honda Insight / Chevrolet Volt systems. (3) 42 V system. (4) GM EVI, Nissan Leaf and Mitsubishi i-MiEV. (5) Heavy-duty hybrid and fuel cell vehicles.